

torch.bernoulli#

<https://pytorch.org/docs/stable/generated/torch.bernoulli.html>

Description:

Draws binary random numbers (0 or 1) from a Bernoulli distribution. The input tensor should be a tensor containing probabilities to be used for drawing the binary random number. Hence, all values in input have to be in the range: $0 \leq \text{input}_i \leq 1$. The i^{th} element of the output tensor will draw a value according to the i^{th} probability value given in input. The returned out tensor only has values 0 or 1 and is of the same shape as input. out can have integral dtype, but input must have floating point dtype.

Function Signature

```
torch.bernoulli(input: Tensor, *, generator: Optional[Generator], out: Optional[Tensor]) → Tensor#
```

Parameters

Keyword Arguments

Parameters

Keyword

generator (torch.Generator, optional) – a pseudorandom number generator for sampling out (Tensor, optional) – the output tensor.

Code Examples

```
>>> a = torch.empty(3, 3).uniform_(0, 1) # generate a uniform
random matrix with range [0, 1]
>>> a
tensor([[ 0.1737,  0.0950,  0.3609],
        [ 0.7148,  0.0289,  0.2676],
        [ 0.9456,  0.8937,  0.7202]])
>>> torch.bernoulli(a)
tensor([[ 1.,  0.,  0.],
        [ 0.,  0.,  0.],
        [ 1.,  1.,  1.]])

>>> a = torch.ones(3, 3) # probability of drawing "1" is 1
>>> torch.bernoulli(a)
tensor([[ 1.,  1.,  1.],
        [ 1.,  1.,  1.],
        [ 1.,  1.,  1.]])

>>> a = torch.zeros(3, 3) # probability of drawing "1" is 0
>>> torch.bernoulli(a)
tensor([[ 0.,  0.,  0.],
        [ 0.,  0.,  0.],
        [ 0.,  0.,  0.]])
```

Detailed Documentation

Documentation

Draws binary random numbers (0 or 1) from a Bernoulli distribution.

The input tensor should be a tensor containing probabilities to be used for drawing the binary random number. Hence, all values in input have to be in the range: $0 \leq \text{input}_i \leq 1$.

The i^{th} element of the output tensor will draw a value according to the i^{th} probability value given in input.

The returned out tensor only has values 0 or 1 and is of the same shape as input.

out can have integral dtype, but input must have floating point dtype.

input (Tensor) – the input tensor of probability values for the Bernoulli distribution

generator (torch.Generator, optional) – a pseudorandom number generator for sampling

out (Tensor, optional) – the output tensor.